AVERAGE POOLING Layer (Take Average Value)

It is a new Module (Average-POOLING module)

Output = 1/r c * Summation (each_input_pixel * 1)

Required to define its input, its output, and Gradient of Output w.r.t input

- Run a sliding window of size $[h_f , w_f]$
- At each location keep the **average** value
- Activation function: $a_{r,c} = \frac{1}{r \cdot c} \sum_{i,j \in \Omega(r,c)} x_{ij}$
- Gradient w.r.t. input $\frac{\partial a_{r,c}}{\partial x_{ij}} = \frac{1}{r \cdot c}$

Where:

r	*Row*
c	*Column*
f	*filter*
h	*Height*
w	*width*

